Leveraging Corporate Engagement for Diversity in Free/Libre and Open Source Software Projects

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***Abstract*—In this workshop paper, we describe results from a mixed-methods study of social diversity and corporate engagement in free/libre and open source software (FLOSS) projects. We collected survey responses from contributors to FLOSS development on GitHub to characterize perceptions of social diversity and corporate involvement in projects. We additionally analyzed data extracted from FLOSS projects hosted on GitHub to investigate differences in diversity based on corporate engagement levels. Our results suggest that organizational decisions may be detrimental to both the expansion of a project's contributor base and for increasing diversity across FLOSS ecosystems. However, as some survey participants note in their responses, organizations subsidizing FLOSS development have opportunities to increase access to and openness of projects which would be beneficial for diversity. This research thus serves to identify organizational factors and actions which harm and help initiatives to improve inclusivity and equity in FLOSS development.**

***Index Terms*—open source software, free/libre software, diversity, inclusion, gender equality, intersectionality, social computing**

I. Social Diversity in FLOSS Development

Diversity in software engineering and in FLOSS development in particular continues to be an area of interest as communities and organizations struggle to make significant progress towards creating inclusive and equitable work spaces. Much of the research on diversity in this work domain focuses largely on gender and/or reduces social identity to a single mode of classification (e.g., either gender or race). This is due, at least in part, to empirical evidence of a large gender imbalance in software development projects. However, recent research describes the issues associated with only considering one facet of social identity and aims to better capture the intersectional nature of experience and injustice [1]. The foundation of this work is based on the concepts of intersectionality [2] and multiple jeopardy [3] which capture the complexity of identity to account for the interdependence of systems used to dominate and discriminate against marginalized groups.

In calls to improve inclusivity in FLOSS development, researchers have suggested that the increased engagement of corporations in FLOSS projects may potentially be leveraged for greater good. Germonprez and colleagues [4] contend that the transformation of this form of software development has far reaching effects "as the change in social construction is accelerating" [4, p. 4], influencing labor in FLOSS projects and in distributed work domains in general. Furthermore, they prompt researchers to consider shifting hierarchies in FLOSS development and the potential effects of the aforementioned transformation on diversity, inclusivity, and equity. These are topical issues for FLOSS development that have implications for both participation and production in technological development. According to the situated knowledge thesis put forth by Donna Haraway [5], "what one knows or experiences reflects one's social, cultural, and historical location" [6, p. 245]. As such, the inclusion of groups that have been traditionally excluded from software development has implications for broader societal issues and labor questions with respect to inclusivity and equity in knowledge production, accessibility and usability of technology, and the direction and application of technological advances. Research in this area has demonstrated that individuals who differ along social dimensions (e.g., gender) also differ along other dimensions that influence problem-solving processes and task outcomes in software development (e.g., information processing style [7]). The effects of varying group composition in software development are evident in the many cases that software has been unusable for, or otherwise failed, some populations [8].

Understanding the need for greater inclusivity and equity in the development of technology, and recognizing the potential of joint efforts between organizations and communities, we designed a mixed-methods study which aims to elucidate the relationship between corporate engagement and social diversity in FLOSS projects. Our research includes both survey data and GitHub project data which we integrate towards this goal. In the next section, we summarize research related to corporate engagement and group imbalances in FLOSS to situate the present study. We then describe our approach and contribution to this research area before moving on to describe our results and identify opportunities for studying and increasing equity in FLOSS development. Through this workshop, we hope to engage in discussion with other participants about the role of organizations, public and private, in promoting inclusivity and equity in FLOSS development. Furthermore, we are interested in building relationships and collaborating with software engineering researchers and practitioners who can provide guidance on how to most effectively integrate our findings in research programs and FLOSS projects.

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II. Related Work

Corporations initially rejected the free and open source paradigm, but, since the mid-2000s, they have maintained a relationship with FLOSS communities, altering development processes and licensing along the way [9]. Different types of corporation-community relationships have been studied in the context of FLOSS development. The findings suggest that a corporation's management style can introduce problems with FLOSS communities. In early research, Dahlander and Magnusson [10] proposed three types of firm-community relationships that fall along a continuum of community benefits: symbiotic (firm gains-community gains), commensalistic (firm gains-community indifferent), and parasitic (firm gains-community loses). In the FLOSS literature, research continues to examine the relationship between communities and the corporations that collaborate with them [11], and the observed and potential effects of the increased presence of corporations in FLOSS development [4]. In such works, the levels at which corporations engage with FLOSS development have been elaborated: economic and business markets; communal resources; "social and material rules" in communities [12]. Ultimately, collaborations between organizations and communities have resulted in a shift away from the development of standard platforms to a focus on usability and features [11], but it is unclear if and to what extent corporations have constrained or facilitated positive social outcomes (e.g., the production of more equitable technologies or inclusive communities [13]).

Germonprez et al. [4] speculate that social computing technologies may hold a space of particular importance in addressing problems emerging in this work domain. Social computing technologies are a type of collaboration infrastructure that is created and designed by organizations, including, for example, social coding platforms like GitHub. Given the embedded nature of these types of technologies in software development, they are critical area for understanding individual, group, and organizational behavior relevant to diversity and inclusivity. Research on diversity and inclusivity in FLOSS projects has largely focused on (1) the presence, or lack thereof, of women; (2) the tenure of women in projects and online platforms; (3) the biases experienced by women and observable in behavioral trace data extracted from version control systems and online platforms; and (4) the experiences and behaviors of women in FLOSS development spaces. Generally, it is observed that women participate in FLOSS projects at lower rates than men [14], [15] and have shorter tenure in projects and online platforms [16], [17]. Some research has begun to explore the presence and effects of bias experienced in FLOSS projects. This work is insightful given that FLOSS contributors have some level of awareness of potential collaborators' gender in social coding platforms [15] and the salience of gender can influence interactions in social spaces [18].

The rate at which the work of women tends to be accepted varies on the basis of both the inferability of their gender and their membership status in FLOSS projects [19]. When a woman's gender is salient and she is not an "insider", her contribution is less likely to be accepted in comparison to a similar contribution made by a male peer whose gender is salient and not an insider. However, this relationship between gender and contribution integration is flipped in the opposite case: a woman whose gender is not identifiable is more likely to have her work accepted when compared to a peer who is a man. Such a finding suggests that the difference in contribution evaluation may be the result of bias rather than level of competence. Other research on the effects of gender bias shows that women tend to contribute to a smaller set of projects relative to men and are more reserved in communication [20] . Women themselves report experiencing discomfort and harassment in FLOSS projects [21] and, perhaps unsurprisingly, gender diversity is relatively uncommon in FLOSS development [15]. Together, these finds show that the interplay of social (e.g., norms and bias) and technical factors (e.g., tenure and experience) in FLOSS projects are associated with gender imbalance.

A recent study suggests that contributors whose race is perceptible and underrepresented tend to have less work accepted in FLOSS projects [22] and at least one survey of developers revealed that conflicts emerge as a result of differences in language, nationality, and ideology between collaborators [23]. Less work has accounted for variation in socioeconomic status, education, and age in FLOSS development participation and experiences. Turning to group and organizational factors, there is some evidence that volunteers' decision to participate in and commitment to a FLOSS project is influenced by social norms and its association with positive causes [24]. At the group level, different types of diversity, namely tenure and gender, have been empirically linked to turnover in FLOSS projects [15]. But, to date, the relationship between organizations, participation decisions, and diversity in FLOSS projects has received little, if any, attention from researchers. We see this gap as opportunity to identify potential avenues and intervention for improving inclusivity and equity in FLOSS development. We now turn to our study of corporate engagement and diversity in FLOSS projects on GitHub.

III. Research Methods

In this section, we describe our research goals and the approaches taken to address them. Our research was motivated by an interest in elucidating the relationship between corporate engagement and diversity in FLOSS projects. More specifically, our goal is to characterize FLOSS contributors' perceptions of corporations as they relate to participation and diversity to better understand if and how these organizations may be leveraged to improve inclusivity and equity. We therefore solicited information about corporations' involvement in FLOSS development from project contributors. With the data collected, we employed a cross-sectional approach to examine differences in responses based on sociodemographic information in addition to the types of projects in which our participants make contributions. In terms of types of projects, we asked participants to specify the level of corporate involvement across all of the projects they contributed to in

the past and at the time of data collection. We complemented our survey research with an analysis of diversity and corporate involvement in FLOSS projects hosted on GitHub. This mixed-methods approach enabled us to provide a more complete picture of study concepts by capturing both individual and group factors, and the differences and similarities between different types of projects. Due to space limitations, we report only a subset of our results. This study was thus completed in two parts–survey research and GitHub project data research–and we organize the text in this section accordingly.

A. Research Questions

Our research was guided by the following questions:

1) **Survey Study**: What are contributors' perceptions of corporate involvement in FLOSS projects?
   a) *Experience*: Do contributor experiences in projects differ based on corporate involvement?
   b) *Participation*: Does corporate involvement influence contributors' intent to sustain participation in a FLOSS project?
   c) *Diversity*: Do contributors believe that corporate involvement helps or hurts the achievement of diversity in FLOSS development?
2) **GitHub Project Analysis**: Are differences in social diversity in FLOSS projects associated with corporate involvement?
   a) *Gender*: Are differences in gender diversity associated with corporate involvement?
   b) *Culture*: Are differences in country diversity associated with corporate involvement?

B. Data Collection

1) Survey Research: We administered a survey via Qualtrics to collect data on contributor perceptions of diversity and corporate involvement in FLOSS projects. The inclusion criterion was contribution to FLOSS development on GitHub. Participants were provided with an explanation of research approved by our institution's Institutional Review Board and asked to provide confirmation of their decision to voluntarily participate in the study. The survey was administered in two phases: a general call for participation distributed via social media and listservs and a targeted invitation to participate based on contribution data extracted from FLOSS projects on GitHub. The first phase distribution resulted in over 400 responses, but as is common with surveys distributed via social media [26], a substantial portion of this data was deemed unusable for analysis. We excluded responses from our data set based on the following criteria: (1) evidence of automated activity, (2) incomprehensible, incoherent, or plagiarized text responses, and (3) incomplete responses. Based on this, we kept 34 complete responses. In the second phase, an invitation to participate was sent directly to the contributors of FLOSS projects selected for the GitHub data component of this research. A total of 596 contributor emails were extracted from commits and pull requests. Of these emails, 10 were returned as undeliverable. This distribution resulted in 57 responses (10% response rate); however, only 30 of these responses were complete and kept for analysis. Our final sample consisted of 64 participants across these two survey distributions.

2) GitHub Data Research: Data were collected and aggregated from the GHTorrent mysql-2019-06-01 data dump for sample selection which included data for over 125 million repos. Additional data were extracted from the GitHub REST API using the PyGitHub library[1] and GrimoireLab's Perceval module[2]. Only repositories that were devoted to software development and were active were classified as belonging to the study population. Specifically, GitHub repos were included in the study population if they met the following criteria: (1) *software development focus*: a specified programming language; (2) *continuity of development activity*: at least 5 years old; (3) *sufficient amount of development activity*: at least 50 commits; (4) *sufficient number of contributors*: at least 5 members. Based on this, the study population was made up of 2,572 repositories—approximately 0.002% of all repos in the GHTorrent data dump.

Before carrying out our analyses, we prepared the data based on best practices for studying FLOSS projects and enhanced individual contributor information. As in prior research on GitHub (e.g., [**?**]), bots were identified and removed from the data set based on their username, information in their GitHub profile bio, and/or GitHub's contributor type label (user or bot). Usernames were used to extract location information provided in GitHub profiles. Location information was prepared for analysis using the tmaptools and tidygeocoder packages in R. Identity merging was applied using an approach based on heuristics described by Vasilescu et al. [15]. As in prior research on gender in GitHub, the genderComputer tool[3] was used to infer the gender of project contributors based on name and location data.

To select a sample of projects for analysis, we aggregated project data across a set of dimensions. This set of dimensions was selected on the basis of their relevance to the research topic [27] and prior research on project growth and attraction of newcomers [28]. For empirical research on participation in FLOSS projects on GitHub, the space consists of the following dimensions: number of contributors (12-month period), number of commits (12-month period), number of forks, number of watchers, main programming language, project age, and project turnover (comparison of two consecutive 12-month periods). These aggregated data then served as input for a sample coverage algorithm [27] to select 20 GitHub projects. Two projects were removed from the sample: one is a mirror of a repo hosted outside of GitHub and the other had been deleted by the time of data collection. The remaining 18 projects were kept for analysis. We then expanded this data set to additionally consider other projects that constituted the ecosystems of the 18 projects. We applied the reference

[1] https://pypi.org/project/PyGithub/
[2] https://github.com/chaoss/grimoirelab-perceval
[3] https://github.com/tue-mdse/genderComputer

TABLE I
GENDER IN SURVEY SAMPLE

Response	N_{social}	$N_{committers}$	Overall (%)
Man	30	24	54 (84%)
Woman	3	1	4 (6%)
Prefer not to say	1	4	5 (8%)
Nonbinary	0	1	1 (2%)

TABLE II
DESCRIPTIVE STATISTICS FOR SURVEY PARTICIPANT AGE

Age	N_{social}	$N_{committers}$
Minimum	21	20
Maximum	44	52
Mean (±SD)	29.51 (±5.81)	34.13 (±8.95)
Median	28	31.50

coupling method proposed by Blincoe and Damian [29] to identify projects in the sample's ecosystems. Our final data set included information for 38 projects, 10,109 contributors, and 44,5217 events (commits, pull requests, and issues).

C. Data Analysis

1) Survey Research: To analyze participant responses to Likert-type items, aggregation and visualization techniques were applied to the data using R and in particular the tidyverse and likert packages. Reflexive thematic analysis (RTA), a contemporary approach for thematic analysis described by Braun and Clark, was applied to participant responses to open-ended survey questions to characterize participants' experiences and dispositions regarding study concepts. This approach is flexible and acknowledges the researcher's "active role in knowledge production" [30, p. 1393].

2) GitHub Projects: As a first step, we supplemented the data with information from READMEs, project websites, and Wiki pages to categorize projects into one of three groups: no corporate influence, receives corporate support, owned by a corporation. Social diversity was then computed using the Blau index [**?**]. Differences between project types were analyzed with statistics and distribution plots.

IV. RESULTS

A. Survey Research

To answer our first set of research questions, we characterize contributors' perceptions of corporations, specifically as they relate to experiences, participation decisions, and effects on diversity (Figures 1-5). When asked about differences between projects based on corporate involvement, most survey

TABLE III
GEOGRAPHIC DISTRIBUTION OF SURVEY PARTICIPANTS

Continent	N_{social}	$N_{committers}$	Overall (%)
Americas	31	12	43 (67%)
Asia	1	4	5 (8%)
Europe	2	13	15 (23%)
Oceania	0	1	1 (2%)

TABLE IV
RACIAL/ETHNIC MAKEUP OF SURVEY SAMPLE

Race/Ethnicity	N_{social}	$N_{committers}$
Arab, Non-White	0	1
Asian	5	3
Black	4	0
Hispanic/Latine, Non-White	0	2
North African	0	1
White, Arab	0	1
White, Hispanic/Latine	1	2
White, Non-Hispanic/Latine	24	19
White, Pacific Islander	0	1

Fig. 1. Gender and experiences in projects with and without corporate involvement.

participants framed their experiences as positive and/or negative, and a small number reported no difference. Positive aspects of projects with corporate involvement reflect personal benefits—career opportunities and compensation—in addition to satisfactory work support processes and characteristics. FLOSS projects provide early-career professionals with the opportunity to build relationships for networking purposes but also for gaining familiarity with specific developers and tasks. Another important component of corporate involvement for FLOSS contributors is the potential for compensation.

Negative aspects of corporate involvement primarily reflected issues with structure, goals, and risk. The structure of corporate-involved projects was viewed as controlling and cumbersome, resulting in an inflexibility around completing taskwork due to top-down constraints. Further, there is a need

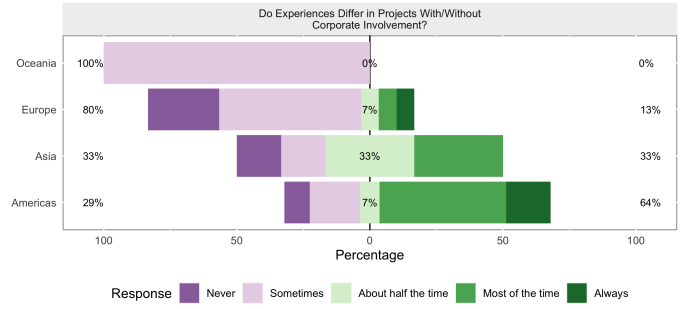

Fig. 2. Geographic region and experiences in projects with and without corporate involvement.

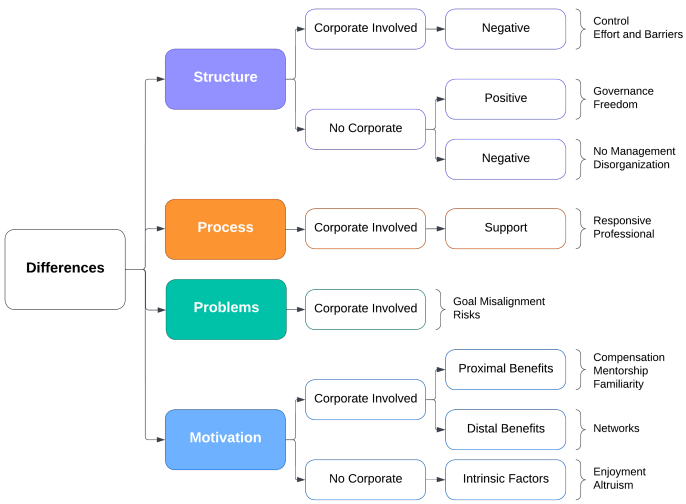

Fig. 3. Themes in participant responses regarding differences between projects with and without corporate involvement.

for increased effort on the part of contributors to determine appropriate processes and meet bureaucratic demands. Participants stated that this extended the time to complete tasks by imposing unnecessary requirements and were likely the result of the software not being a priority for the associated company. As one participant noted: *"most of the time, projects with corporate involvement tend to lead to a lower quality of work and more time consumed as corporations set up initiatives and guidelines we all have to follow which slows everything down"* [SM2]. While corporations may then be associated with faster times with respect to user support, they are also associated with increased time and effort on the part of code contributors with no perceived benefit to software quality.

Other factors which produced distinct experiences and influenced participation include the goals of companies involved in FLOSS and potential risks of joining projects with corporate involvement. A set of risks identified by survey participants exists at the project level: abandonment, corporate wrongdoing, scale, and conflict. These first two risks were seen as potentially producing the same effect on project success: *"Because a corporation may be involved in a FLOSS project, there is a concern that if the corporation in question is involved in nefarious dealings (or voluntarily abandons support) that it can severely damage or limit the momentum of a project"* [GH6]. The third type of risk similarly reflects a concern about insufficient resources, but in the opposite case: a FLOSS project with corporate involvement may become popular and grow significantly without the personnel or infrastructure to meet needs at scale.

Finally, the fourth risk is linked to control and the foundations of FLOSS, namely licensing, in which participants expressed concerns about legal issues that may emerge from contribution to projects with corporate involvement. In particular, these concerns centered on the possibility that their contributions would become restricted as the intellectual property of the company in question. Each of these risks leads some FLOSS contributors to exhibit hesitancy in joining or completely avoid corporate-backed projects. However, several participants described the importance of familiarity with a company's reputation in FLOSS communities as this better enabled the assessment of risk for participation decisions.

Although fewer participants focused on the positive and negative aspects of FLOSS projects without corporate involvement, those who did described the lack of structure as producing benefits and consequences. On the positive side, the governance models of non-corporate projects are perceived as fairer and granting greater freedom to contributors. This allows them to improvise to produce code that addresses their own and others' needs. On the negative side, the lack of structure can create coordination issues: *"projects without corporate involvement are more public, but also more disorganized, with mostly no one doing project management"* [GH4].

Survey participants detailed a number of reasons they might choose to end or continue their participation in corporate-involved projects. In general, participants' responses reflected the differences they experienced between these types of projects. Among reasons to leave a project, participants expressed that they experienced discord with decision makers and a misalignment of values. While some were motivated to continue due to perceived personal benefits, others were demotivated by a lack of collective need and social good: *"I'm less interested in helping with projects that already have corporate funding since they have the money to hire someone, while other projects I could work on don't"* [SM15] and *"It's more of a moral dilemma - Like why should I contribute to the benefit of a corporate whose only motivation is to become richer? I would rather spend my time with FOSS projects whose contributors are driven by passion for programming (most of them) and the prime motivation being to create better software and make everyone's life a bit better"* [SM28]. These participants affirmed their commitment to the communities and values of free software in their contribution decisions. Still, some participants clarified that they were more nuanced in evaluating projects: their decisions were informed by how the contribution experience felt or based on an established history between the corporation and FLOSS communities. Even in situationally informed decisions, contributors expressed a limit to such nuance as some entities were to be avoided at all costs: *"I [am] sensitive to a small set of organisations with whom I do not want any association"* [SM21].

The results of this survey research provide evidence for the effect of corporate activities and decisions on individual decision making related to participation in FLOSS development. Participants in the sample not only confirmed that there are key differences in individual preferences, experiences, and beliefs with respect to project types, they provided a richer understanding of their perceptions through descriptions of the rationale which informs their perspective and in turn guides their decision making. While they varied in their experiences, survey participants generally held neutral or positive views of corporations effects on diversity in FLOSS projects (Figures 4

and 5). We next describe the results of a complementary approach to our survey research: an analysis of big data extracted from FLOSS project ecosystems in GitHub.

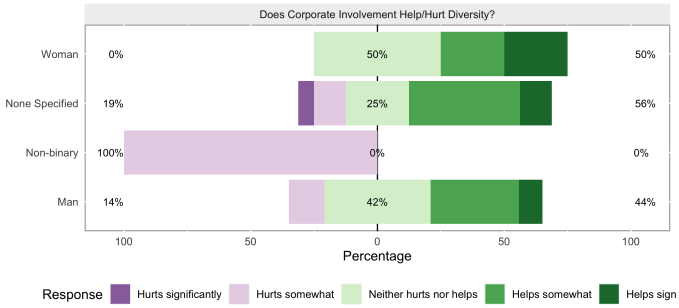

Fig. 4. Differences between projects with and without corporate involvement.

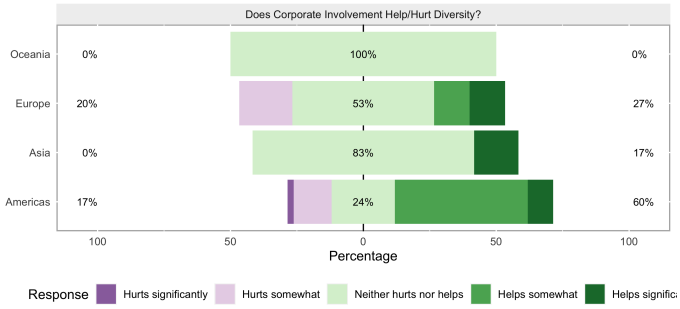

Fig. 5. Differences between projects with and without corporate involvement.

B. GitHub Data Research

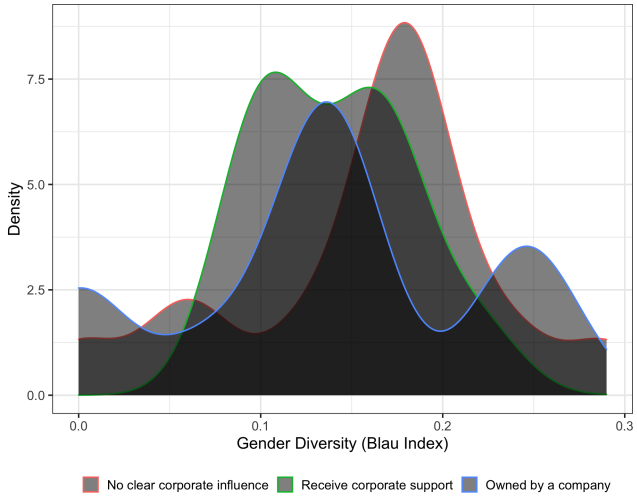

Fig. 6. The distribution of gender diversity by project type.

To answer our second set of research questions, we describe the results of a quantitative analysis of differences in diversity based on levels of corporate involvement in FLOSS development. Gender diversity is generally low in projects (<0.3) while country diversity is high (>0.75). A Welch's unequal variances *t*-test, a modification of the two sample *t*-test that is appropriate for cases in which there are unequal groups and there is unequal variance between those groups, was applied to the data. No significant difference was found between the two project types of interest, $t(22) = 0.74$, $p = .47$, d = 0.25 (lower CI: -0.51; upper CI: 1.01), with no corporate influence projects ($M = 0.16$, $SD = 0.07$) having, on average, similar levels of gender diversity as company-owned projects ($M = 0.14$, $SD = 0.09$). The skewness and prevalence of ties between project types limited the application of statistical tests to country diversity data.

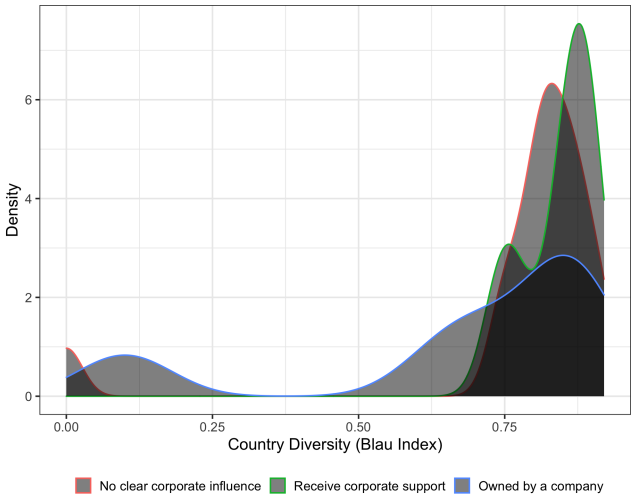

Fig. 7. The distribution of country diversity by project type.

V. Discussion

The goal of this research was to characterize the perceptions of diversity and corporations held by FLOSS project contributors and to examine if and how this is reflected in observable behavior in a social coding platform. Relevant to our first research question, the results of the survey highlight the heterogeneity of FLOSS contributors' beliefs, behaviors, and experiences. Contributors report experiencing important differences between projects based on corporate involvement (*RQ1a*). They also vary in the degree to which their participation decisions are influenced by the presence of corporations (*RQ1b*) and perceive corporations are helping or harming diversity (*RQ1c*). The results of the GitHub data analysis indicate that company-owned FLOSS projects are in general neither significantly better nor worse than community-maintained projects in terms of gender diversity (*RQ2a*) but may be less open to a diverse set of potential contributors when looking at categories other than gender like culture (*RQ2b*).

A key takeaway for this research is that FLOSS contributors experience important differences between projects based on the presence of corporations in them. When looking at behavioral data extracted from FLOSS projects, we see that the presence of a corporation or company is not associated

with greater levels of diversity, even though some contributors perceive corporations as having a positive influence on this dimension of group composition. Organizations interested in affecting change in FLOSS projects must reflect on why their own internal practices, norms, and/or culture have not already produced diverse collaborations. Meaningful action and policy relevant to internal dynamics is paramount before any attempt to impose top-down measures on FLOSS communities, which display sensitivity to control, is made in projects. This is essential for not only influencing those communities, but also ensuring the integrity and longevity of diversity as a commitment. Communities of contributors who are especially oriented towards software freedom rather than personal benefits and profit are unlikely to respond positively when they perceive organizational actions as undue or unjustified control. Surface-level actions without meaningful change to organizational policy are seen as empty gestures centered on presenting an image of inclusivity and concern for equity rather than an embodiment of such values. As one survey participant expressed, *"I'm displeased by GitHub's move to rename the "master" branch, I think it's a bullshit statement and it doesn't solve any of our modern-day problems"* [GH11]. Research on diversity and anti-discrimination programs in the workplace have shown that such initiatives, motivated by a desire to protect public image and reduce legal risks, do not only fail to increase diversity, they effectively reduce it further [31].

The results of the GitHub data analysis suggest that company-owned projects tend to have lower levels of country diversity. Even though country diversity was generally high across all project types, it was limited in certain continents and geographical regions—Africa, Oceania, West and South Asia, and South America had lower levels of representation compared to North America, Europe, and East Asia. Organizations interested in extending use of FLOSS and participation in projects to a broader group may then want to consider ways that they can lower barriers to contribution for these underrepresented populations. A survey participant provided an example of a relevant barrier that limits the achievement of diversity: *"While open source is used worldwide, not everyone has the technical ability or time to contribute. Being able to measure diversity in projects, and work to improve it, would be a huge benefit to the broader open source community. For example, much code is written by English-speaking programmers, making it challenging for non-native speakers to follow comments, method / function names, etc."* [SM34]. Addressing such barriers can then help move FLOSS towards greater inclusivity at the global level. In this specific case, organizations can intentionally devote resources to translation, allocating funds to services and paying non-English speakers in underrepresented countries to support the completion of this work. Such an action would not only immediately confer benefits to global labor, it could potentially have cascading effects by providing access to a much broader audience.

The results of the RTA applied to participant responses regarding their experiences and feelings towards corporations in FLOSS development reveal that underlying motivational and reward differences for working in varied contexts. In the first case, a subset of FLOSS contributors delineate the role of their economic needs and management style preferences. These participants note that their contributions to corporate-involved projects grant them immediate and more distal benefits in the form of compensation and networks, respectively. They additionally note that the management styles employed in projects with corporate involvement ensure that they have access to personnel and resources to enable the completion of their work.

In the second case, a subset of FLOSS contributors describe the impediments of management style to their work, making it needlessly laborious through increased bureaucracy and constraining their ability to innovate due to top-down control. Further, they may view the profit-based goals of the corporation to not be in line with their personal values—improving codebases as a greater good—or the broader values of openness and unrestricted use in the free software development paradigm. This extends prior work in the organizational sciences by elaborating the specific economic, organizational, and work-related factors that drive observed membership change in technology development, and provides evidence for a set of FLOSS contributor types that can be differentiated along these three dimensions.

The importance of these dimensions and contributor types is more salient when connecting them to the development of interventions to improve diversity and inclusion in FLOSS. The former group appears amenable to top-down interventions under conditions where they are able to continue to derive economic benefits and resources to support this work remain available. The latter group in contrast may reject top-down interventions, particularly when they lack transparency and leadership fails to make explicit how such interventions will produce benefits for code quality. The findings presented here can inform future research to test and develop interventions in FLOSS in order to make advances in this important contemporary form of work.

The adoption of FLOSS development practices by organizations and the growing number of collaborative multi-organizational/institutional collaborative initiatives may indeed be useful in encouraging a more inclusive, global workforce. Yet, organizations invested in making change must practice care in pursuing such collaborations. In particular, the responses of survey participants suggest that some corporations lack social capital with respect to FLOSS communities (contributors and users). Therefore, it is important for organizations to be judicious in forming collaborations. Further, they should heed the advice given to those seeking to participate in FLOSS development: thoroughly vet the history of potential collaborators with affected communities.

Acquisitions in FLOSS ecosystems can also be quite disruptive depending on the social-normative attitudes held by the FLOSS community and may produce a loss with respect to advances made towards greater inclusivity and diversity. For example, Microsoft has a contentious relationship with FLOSS communities, being an early opponent of free and

open source licensing. Following Microsoft's acquisition of GitHub, some survey participants shared that they migrated away from the platform, while others braced for an inevitable departure, taking a 'wait and see' attitude as to when corporate ownership would impinge on their values.

VI. Conclusion

We examined differences in participation diversity between projects with different organizational structures and control mechanisms, showing that they vary in terms of both perceived and actual openness and inclusivity. The qualitative component of this research revealed subgroups among FLOSS contributors in terms of values and goals: study participants report that they are guided by a desire to reap personal benefits or maintain codebases as a social good. The results of the quantitative component research suggest that variations in participation occur on the basis of corporate involvement. While there was no evidence of differences along the dimension of gender, some evidence was found that projects with no corporate influence exhibit higher levels of openness along the dimension of geographical location as they had among the highest levels of country diversity. This study suggests that to differentially influence participation and both encourage and maintain diversity in FLOSS projects, it is important to develop strategies that are appropriate for contributor base.

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